Lecture 22: MA Models and Forecasting

1 Moving Average Models

The general form of a $MA(q)$ model is

$$r_t = c_0 + a_t - \sum_{i=1}^{q} \theta_i a_{t-i} \tag{1}$$

It can be seen easily that the unconditional mean is $\mathbb{E}[r_t] = c_0$, and the unconditional variance is

$$\gamma_0 := Var(r_t) = \left(1 + \sum_{i=1}^{q} \theta_i^2\right) \sigma_a^2.$$

Moving-average models are always weakly stationary, because they are finite linear combinations of a white noise sequence, for which the first two moments are time invariant.

Let us next compute the auto-covariance function of the model. For $l > q$, $\gamma_l := cov(r_t, r_{t-l}) = 0$. For $1 \leq l \leq q$,

$$\begin{aligned}
\gamma_l &= cov(r_t, r_{t-l}) = -\sum_{i=l}^{q} \theta_i cov(a_{t-i}, r_{t-l}) \\
&= -\sum_{i=l}^{q} \theta_i cov\left(a_{t-i}, a_{t-l} - \sum_{j=1}^{q} \theta_j a_{t-l-j}\right) \\
&= \left[-\theta_l + \sum_{i=l+1}^{q} \theta_i \theta_{i-l}\right] \sigma_a^2
\end{aligned}$$

Recall that the lag-l autocorrelation is given by $\rho_l = \frac{\gamma_l}{\gamma_0}$. This yields

$$\rho_l = \frac{\gamma_l}{\gamma_0} = \frac{-\theta_l + \sum_{i=l+1}^{q} \theta_i \theta_{i-l}}{1 + \sum_{i=1}^{q} \theta_i^2}$$

Forecasting: Assume the time series follows $MA(q)$. The l-step ahead forecast is given by

$$\hat{r}_T(l) := \mathbb{E}\left[r_{T+l} \middle| \mathcal{F}_T\right] = c_0 - \sum_{i=l}^{q} \theta_i a_{T+l-i}.$$

The forecasting error is given by

$$\hat{e}_T(l) := r_{T+l} - \hat{r}_T(l) = a_{T+l} - \sum_{i=1}^{\min(l-1,q)} \theta_i a_{T+l-i}.$$

The variance of the forecasting error is then given by

$$\mathbb{E}\left[\left(\hat{e}_T(l)\right)^2\right] = Var\left(\hat{e}_T(l)\right) = \sigma_a^2 \left(1 + \sum_{i=1}^{\min(l-1,q)} \theta_i^2\right).$$

2 Estimation of MA(q) models

The maximum likelihood method is typically used to estimate a $MA(q)$ model. Suppose r_t, $t = 1, \ldots T$, are observed. We consider the conditional likelihood function of r_t, $t = 1, \ldots, T$, given $a_0, a_{-1}, \ldots, a_{-q+1}$. For each $t = 1, \ldots, T$, we have

$$a_t = r_t + \sum_{i=1}^{q} \theta_i a_{t-i} - c_0.$$

Given $a_{t-1}, \ldots, a_{t-q}$, we have that r_t follows a Gaussian with mean $c_0 - \sum_{i=1}^{q} \theta_i a_{t-i}$ and variance σ_a^2. The conditional log-likelihood of $r_1, \ldots, r_T$ is given by

$$\begin{aligned} \log L(r_{1:T} \big| c_0, \theta_{1:q}, a_{-q+1:0}) &= -\frac{T}{2}\log(2\pi) - \frac{T}{2}\log\sigma_a^2 \\ &\quad -\frac{1}{2\sigma_a^2}\sum_{i=1}^{T}\left(r_t - \left(c_0 - \sum_{i=1}^{q}\theta_i a_{t-i}\right)\right)^2 \end{aligned}$$

2.1 Example

We present an application to estimation of bid-ask spreads in financial time series. The example is based on the paper R. Roll (1984), A Simple Implicit Measure of the Effective Bid-Ask Spread in an Efficient Market, Journal of Finance. Suppose that P_t^* is the unobserved fundamental price of the asset without the bid-ask spread. Moreover, assume that $\eta_t := P_t^\star - P_{t-1}^\star$ are i.i.d. with zero mean and variance σ_η^2. Let $s > 0$ be the bid-ask spread. Then the observed asset price is given by

$$P_t = P_t^\star + \frac{s}{2} I_t,$$

where $I_t = 1$ if it is a buyer initiated transaction (the buyer would have to buy the asset at the ask price given by $P_t = P_t^\star + s/2$). Instead, $I_t = -1$ if it is a seller initiated transaction (the seller would have to see the asset at the bid price given by $P_t = P_t^\star - s/2$). Assume that I_t's are i.i.d. and have zero mean, i.e., $P(I_t = 1) = P(I_t = -1) = \frac{1}{2}$. This implies $Var(I_t) = 1$ for each t.

Consider the observed stock price change given by

$$y_t := P_t - P_{t-1} = \eta_t + \frac{s}{2}\left(I_t - I_{t-1}\right).$$

The variance is $Var(y_t) = \sigma_\eta^2 + \frac{s^2}{2}$. The auto-covariance is given by

$$\gamma_1 := cov(y_t, y_{t-1}) = -\frac{s^2}{4}.$$

Moreover, $\gamma_l = 0$ for $l \geq 2$. It then follows directly the the lag-1 autocorrelation is

$$\rho_1 = -\frac{s^2/4}{\sigma_\eta^2 + s^2/2},$$

while $\rho_l = 0$ for $l \geq 2$. Hence, the observed price change y_t follows a time-series resembling a MA(1) model with negative autocorrelation, although the fundamental price change is i.i.d.